

URTE Arcadia / Capella MBSE Extension

SysML Viewpoints and Extended Schemas

Compliant with **Arcadia** Method and **Capella** Tooling

for the

Universal Regeneration Therapy Ecosystem (URTE)

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Extension of URTE MBSE Document (June 2026)

*Based on URTE Architecture Document, Investment Prospectus,
Infrastructure Model, and Situation Room Briefing*

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Abstract

This document extends the URTE MBSE baseline with detailed **Arcadia**-compliant viewpoints and **SysML**-like schemas suitable for direct implementation in the **Capella** open-source MBSE tool.

It applies the full Arcadia method (Operational Analysis, System Analysis, Logical Architecture, Physical Architecture, and EPBS) to the Universal Regeneration Therapy Ecosystem. All schemas are provided in textual/SysML-inspired notation that can be recreated in Capella using standard Arcadia viewpoints, allowing traceability from operational needs through to physical deployment while preserving the scale-invariant, TRL Pull, and ethical-by-design principles of URTE.

Contents

1. Introduction to Arcadia Method for URTE

1.1 Why Arcadia / Capella for URTE?

The URTE system is a complex, multi-scale, socio-technical System-of-Systems involving: - Biological and synthetic components (bionanorobots, cell/gene therapies) - AI-intensive software (digital twins, generative design, RL optimizers) - Physical deployment across clinical, terrestrial, and space environments - Strong ethical, regulatory, and planetary-boundary governance constraints

Arcadia (developed by Thales) provides a structured, viewpoint-based methodology that is particularly well-suited because it emphasizes: - Clear separation between **what** the system must do (Operational System Analysis) and **how** it is realized (Logical Physical Architecture) - Strong support for **functional chains** and **data flows** across scales — critical for URTE’s cross-scale coupling - Built-in traceability and impact analysis — essential for TRL Pull mechanism and regulatory compliance - Native support in the free **Capella** tool, enabling collaborative MBSE without heavy licensing costs

This extension document maps the existing URTE 6-layer architecture and biomathematical models into standard Arcadia viewpoints, providing ready-to-use schemas for Capella implementation.

1.2 Arcadia Viewpoint Structure Applied to URTE

Arcadia Viewpoint	URTE Focus & Key Artifacts
Operational Analysis (OA)	Operational Capabilities (e.g., “Regenerate damaged tissue at subcellular scale”, “Restore ecosystem resilience at landscape scale”), Operational Entities (Clinicians, Ecologists, Space Agencies, Ethics Boards), Operational Activities and Scenarios
System Analysis (SA)	System Capabilities, System Functions (allocated from the 6 layers), Functional Chains (Intervention Lifecycle, TRL Pull Flywheel, Cross-Scale Learning), System Actors, Data Model (multi-scale state vectors u_s)
Logical Architecture (LA)	Logical Components (the 6 URTE layers as reusable Logical Components with Ports & Interfaces), Logical Functions, Logical Data Flows, Allocation of Functions to Components
Physical Architecture (PA)	Physical Nodes (Ground Mission Control, Clinical Deployment Hubs, Field Kits, MedInt Satellite Constellation, Orbital/Lunar Precursors), Physical Components, Deployment of Logical Components, Physical Links
EPBS	Configuration Items, Versioned Builds, Safety/Security Classification per Component

2. Operational Analysis (OA) Viewpoint

2.1 Operational Capabilities

- OC-1 Regenerate living systems at molecular/cellular scale with precision and safety.
- OC-2 Restore ecosystem functions and resilience at landscape/regional scale.
- OC-3 Enable controlled planetary regeneration precursors and bio-regen + ISRU for space habitats.
- OC-4 Accelerate technology maturation across domains via TRL Pull mechanism.
- OC-5 Enforce ethical, planetary-boundary, and governance constraints on all interventions.

2.2 Operational Entities & Actors

- Clinician / Patient Entity
- Ecologist / Restoration Practitioner
- Space Agency / ISRU Engineer
- Multi-Stakeholder Ethics Board
- Regulatory Authority (FDA/EMA/UN)
- General Public / Civil Society

2.3 Operational Architecture Diagram (Textual SysML-inspired)

Listing 2.1: Operational Architecture (OA) – Textual Representation for Capella

```
1 Operational Architecture [OA]
2   Operational Entities
3     Clinician/Patient
4     Ecologist/Restoration Team
5     Space Agency/ISRU Team
6     Ethics Board (Multi-Stakeholder)
7     Regulatory Authority
8   Operational Activities
9     OA.1 Perform Subcellular Repair
10    OA.2 Execute Landscape Restoration
11    OA.3 Conduct Mars/Moon Analog Campaign
12    OA.4 Authorize High-Impact Intervention (Ethics Gate)
13    OA.5 Monitor Planetary Boundary Indicators
14   Operational Capabilities
15     OC.1 Precision Multi-Scale Regeneration
16     OC.2 TRL Pull Acceleration
17     OC.3 Ethical & Planetary Guardrail Enforcement
18   Operational Scenarios
19     Scenario 1: First-in-Human Bionanorobot Therapy (with sandbox
20     progression)
21     Scenario 2: Cross-Domain Learning -- Clinical success improves
ecosystem model
```

3. System Analysis (SA) Viewpoint

3.1 System Capabilities

- SC-1 SC.1 Provide unified multi-scale digital twin simulation (molecular $\leftrightarrow$ planetary)
- SC-2 SC.2 Generate and optimize intervention protocols under guardrails
- SC-3 SC.3 Execute progressive sandboxed physical interventions
- SC-4 SC.4 Enable closed-loop cross-domain learning and TRL Pull
- SC-5 SC.5 Maintain full traceability, auditability, and ethical oversight

3.2 System Functions (Derived from URTE 6 Layers)

Function ID	System Function Description	Allocated Layer	to
F.1.1	Authorize intervention via multi-stakeholder ethics board	Layer 1	
F.1.2	Enforce planetary-boundary mathematical guardrails	Layer 1 + 2	
F.2.1	Run Physics-Informed Neural Network (PINN) multi-scale simulation	Layer 2	
F.2.2	Generate candidate intervention protocols (generative design)	Layer 2	
F.2.3	Optimize interventions via stochastic optimal control ($J(u)$)	Layer 2	
F.2.4	Perform uncertainty quantification	tipping-point warning	early
Layer 2			
F.3.1	Execute high-TRL clinical regen modules (cell/gene, bioprinting)	Layer 3	
F.4.1	Execute eco-planetary regen protocols	Layer 4	
F.5.1	Deploy bionanorobot swarms with kill-switches	containment	
Layer 5			
F.5.2	Collect real-time telemetry from all scales	Layer 5	
F.6.1	Update digital twin models from outcome data	Layer 6	
F.6.2	Perform cross-domain learning (clinical $\leftrightarrow$ ecological)	Layer 6	

3.3 Key Functional Chain: Intervention Lifecycle with TRL Pull (Activity Diagram - Textual)

Listing 3.1: Functional Chain: End-to-End Intervention Lifecycle + TRL Pull

```

1 Functional Chain: Intervention Lifecycle with TRL Pull (SA)
2 Start
3   --> [F.1.1] Ethics Board Authorization Gate
4   --> [F.2.2] Generative Protocol Designer (Layer 2)
5   --> [F.2.3] RL Optimizer with Guardrails (R, planetary boundaries)
6   --> Decision: High-Impact? --> [F.1.1] Ethics Veto possible
7   --> [F.5.1] Progressive Sandbox Deployment (in-vitro -> animal -> controlled
    pilot)
8   --> [F.5.2] Real-time Telemetry Collection
9   --> [F.6.1] Digital Twin Model Update
10  --> [F.6.2] Cross-Domain Learning --> TRL Pull to lower-maturity modules
11  --> End (Outcome logged + Audit trail complete)

```

3.4 Data Model – Multi-Scale State Vector (SysML Block)

Listing 3.2: Data Model: Multi-Scale State Vector (SysML Block Definition)

```

1 Block: MultiScaleStateVector
2   Attributes:
3     + scaleLevel: ScaleEnum {Molecular, Cellular, Tissue, Organ, Ecosystem,
        Regional, Planetary}
4     + u_s: StateVector           // concentrations, stress, biomass, CO2, etc.
5     + lambda_dom: Real           // dominant eigenvalue for resilience R
6     + variance: Real             // early-warning indicator
7     + lag1_autocorr: Real
8     + skewness: Real
9     + kurtosis: Real
10  Operations:
11    + computeResilience(): Real   // R = -lambda_dom
12    + detectTippingPoint(): Boolean
13  Constraints:
14    {planetaryBoundaryGuardrails enforced on all u_s updates}

```

4. Logical Architecture (LA) Viewpoint

4.1 Logical Components (Mapping URTE 6 Layers)

The six URTE layers are modeled as reusable **Logical Components** with well-defined ports and interfaces. This enables strong modularity and traceability.

Listing 4.1: Logical Architecture (LA) – URTE Layers as Logical Components

```
1 Logical Architecture [LA]
2     Logical Component: CommandAndGovernance (Layer 1)
3     Ports:
4         - EthicsVetoPort (in/out)
5         - RegulatoryInterfacePort
6         - AuditLogPort
7     Allocated Functions: F.1.1, F.1.2
8
9     Logical Component: AIOrchestrationAndMultiScaleSimulation (Layer 2)
10    Ports:
11        - TwinInputPort (sensor data, u_s)
12        - ProtocolOutputPort (candidate interventions)
13        - GuardrailEnforcementPort
14    Allocated Functions: F.2.1 -- F.2.4
15
16    Logical Component: BiologicalRegenSubsystem (Layer 3)
17    Ports:
18        - ClinicalDataPort
19        - TherapyExecutionPort
20    Allocated Functions: F.3.1
21    Note: Primary TRL Pull revenue & data source (TRL 6-9)
22
23    Logical Component: EcoPlanetaryRegenSubsystem (Layer 4)
24    Ports:
25        - EcoTelemetryPort
26        - RestorationProtocolPort
27    Allocated Functions: F.4.1
28
29    Logical Component: SensingActuationAndDeployment (Layer 5)
30    Ports:
31        - TelemetryOutPort
32        - InterventionCommandPort
33        - KillSwitchPort (safety)
34    Allocated Functions: F.5.1, F.5.2
35    Sub-Components:
36        - BionanorobotSwarm (with velocity regime models)
37        - IoTSensorNetwork
38        - BioreactorBioprinterFleet
39
40    Logical Component: ContinuousFeedbackAndLearningLoop (Layer 6)
41    Ports:
42        - ModelUpdatePort
43        - CrossDomainLearningPort
44    Allocated Functions: F.6.1, F.6.2
45    Note: Implements Virtuous Evidence Flywheel + TRL Pull
```


4.2 Logical Internal Block Diagram (IBD) – Simplified Textual

Listing 4.2: Logical IBD: Cross-Layer Data Flow (Arcadia Style)

```
1 Logical IBD: URTE Core Data Flow
2 [CommandAndGovernance] --EthicsVeto--> [AIOrchestration...]
3 [AIOrchestration...] --Protocol--> [SensingActuation...]
4 [SensingActuation...] --Telemetry--> [ContinuousFeedback...]
5 [ContinuousFeedback...] --ModelUpdate--> [AIOrchestration...]
6 [BiologicalRegen] --ClinicalOutcomeData--> [ContinuousFeedback...] // TRL Pull
   source
7 [EcoPlanetaryRegen] --EcoOutcomeData--> [ContinuousFeedback...]
```

5. Physical Architecture (PA) Viewpoint

5.1 Physical Nodes & Deployment

Listing 5.1: Physical Architecture (PA) – Nodes and Deployment

```
1 Physical Architecture [PA]
2   Physical Node: GroundMissionControlCenter
3     Hosts: CommandAndGovernance (Layer 1)
4           AIOrchestrationAndMultiScaleSimulation (Layer 2)
5           ContinuousFeedbackAndLearningLoop (Layer 6)
6     Links: High-bandwidth to Clinical Hubs and Satellite Ground Stations
7
8   Physical Node: ClinicalDeploymentHub
9     Hosts: BiologicalRegenSubsystem (Layer 3) + parts of Layer 5
10    Components: Bioprinters, Robotic Surgical Systems, Local Telemetry Cache
11
12   Physical Node: FieldEcoRestorationKit (mobile/edge)
13     Hosts: parts of EcoPlanetaryRegenSubsystem + SensingActuation
14     Components: Drone Swarms, IoT Gateways, Portable Bioreactors
15
16   Physical Node: MedIntSatelliteConstellation (Ecological Measurement)
17     Hosts: parts of Layer 5 (remote sensing + telemetry relay)
18     Links: to GroundMissionControl and SpaceStationGovernanceNode
19
20   Physical Node: SpaceStationGovernanceNode (Phase 3+)
21     Hosts: Governance oversight for MedInt constellation + precursor bio-regen
22           payloads
23     Links: to Orbital/Lunar Bio-Regen + ISRU Modules
```

5.2 Physical Deployment Diagram (Textual)

Listing 5.2: Physical Deployment: URTE Components to Nodes

```
1 Deployment:
2   CommandAndGovernance      --> GroundMissionControlCenter
3   AIOrchestration...        --> GroundMissionControlCenter
4   BiologicalRegenSubsystem   --> ClinicalDeploymentHub
5   EcoPlanetaryRegenSubsystem --> FieldEcoRestorationKit +
6   GroundMissionControlCenter
7   SensingActuation...        --> ClinicalDeploymentHub + FieldEcoRestorationKit
8   + MedIntSatellites
9   ContinuousFeedback...      --> GroundMissionControlCenter
```

6. EPBS and Configuration Management

End-Product Breakdown Structure (EPBS) for URTE Phase 1 (recommended Capella structure):

- **URTE Core Platform v1.0** (Configuration Item)
 - Digital Twin Engine (PINN-based)
 - AI Orchestration Module
 - Governance & Ethics Dashboard
 - Bionanorobot Sandbox Kit (v0.8)
- **Clinical Pilot Deployment Package**
- **Terrestrial Eco-Restoration Pilot Package**
- **Data Standards & API Library** (open source part)

Each Configuration Item carries safety classification, TRL level, and traceability to requirements.

7. Cross-Viewpoint Traceability & Capella Implementation Guidance

7.1 Traceability Matrix (Excerpt)

Arcadia Element	URTE MBSE Requirement	TRL / Phase
Operational Capability OC.1	FR-01, SR-01	Phase 1–3
Functional Chain "Intervention Lifecycle + TRL Pull"	FR-04, NFR-03	All Phases
Logical Component "AIOrchestration..."	FR-01, FR-02	Phase 1
Physical Node "MedIntSatelliteConstellation"	Phase 3 roadmap	Phase 3

7.2 Recommended Capella Project Structure

1. Create new Capella project: URTE_MBSE_v1
2. Use standard Arcadia viewpoints in order:
 - (a) Operational Analysis
 - (b) System Analysis
 - (c) Logical Architecture
 - (d) Physical Architecture
 - (e) EPBS
3. Define reusable libraries for:
 - MultiScaleStateVector (Data)
 - BionanorobotSwarm (Logical + Physical)
 - GuardrailConstraints (Constraints)
4. Use Capella's built-in traceability matrix and impact analysis views for TRL Pull and regulatory evidence.
5. Export diagrams as images or SVG for inclusion in documentation (or use Capella's HTML/Word export).

8. Conclusion and Next Steps

This Arcadia/Capella extension provides a complete set of viewpoints and textual SysML schemas that faithfully represent the URTE system while being directly implementable in the free Capella tool.

Recommended immediate next steps:

1. Import the Logical Components and Functional Chains into Capella.
2. Create the first version of the multi-scale data model as a shared library.
3. Define initial Operational Scenarios for the four Phase 1 pilots.
4. Run Capella's validation rules to ensure consistency across viewpoints.
5. Generate documentation exports for integration into the main URTE MBSE Document.

The combination of the original URTE MBSE Document + this Arcadia extension gives a full, tool-ready MBSE baseline aligned with both academic best practices (SEBoK/SWEBOK) and industrial methodology (Arcadia/Capella).

— *End of URTE Arcadia / Capella MBSE Extension v1.0* —

A. Quick Reference: Arcadia Viewpoint Mapping to URTE Layers

Arcadia View	URTE Layer Focus	Key SysML Element
Operational Analysis	Stakeholder needs across clinical, eco, space	Operational Capabilities + Scenarios
System Analysis	What the system must achieve (6-layer functions)	System Functions + Functional Chains
Logical Architecture	How the system is structured (reusable components)	Logical Components (6 layers) + Ports
Physical Architecture	Where and how it is deployed (nodes)	Physical Nodes + Deployment
EPBS	Configuration & build management	Configuration Items

B. Selected Extended ASCII SysML Diagrams

B.1 Block Definition Diagram (BDD) – URTE Core Logical Components

```
1 BDD: URTE_Logical_Components
2 block CommandAndGovernance
3 block AIOrchestrationAndMultiScaleSimulation
4 block BiologicalRegenSubsystem
5 block EcoPlanetaryRegenSubsystem
6 block SensingActuationAndDeployment {
7     block BionanorobotSwarm
8 }
9 block ContinuousFeedbackAndLearningLoop
10
11 AIOrchestrationAndMultiScaleSimulation --> BiologicalRegenSubsystem : <<allocate>>
12 AIOrchestrationAndMultiScaleSimulation --> EcoPlanetaryRegenSubsystem : <<allocate
13     >>
14 SensingActuationAndDeployment --> BionanorobotSwarm : composition
```

B.2 State Machine Diagram – Intervention Authorization & Sandbox Progression

```
1 StateMachine: InterventionLifecycle
2 [*] --> EthicsReview
3 EthicsReview --> Authorized : ethicsBoard.approve()
4 EthicsReview --> Rejected : ethicsBoard.reject()
5 Authorized --> InVitroSandbox
6 InVitroSandbox --> SmallAnimalSandbox : validation.pass()
7 InVitroSandbox --> Paused : safetyTrigger()
8 SmallAnimalSandbox --> ControlledPilot : validation.pass()
9 ControlledPilot --> OperationalDeployment : regulatory.approval()
10 OperationalDeployment --> Completed
11 Paused --> EthicsReview : rootCauseResolved()
```